

Exam Date & Time: 22-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

AI for Medical Image Analysis [CSE-4011]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate and discuss the various types of interactions that may arise in image segmentation tasks. (5)
 - A)
 - B) Demonstrate and analyze the significance of "domain knowledge about the objects" in the image segmentation task with an example. (3)
 - C) Predict whether the following statements are true or false and justify your answer. "Segmentation by texture plays a prominent role in medical image segmentation when properties of textured organs are too subtle to be differentiated". (2)
- 2) Analyse in detail the working principle of the "bag of features". Also, highlight how it can be applied to the domain of medical image analysis. (5)
 - A)
 - B) Interpret in detail the working principle of MSER features. (3)
 - C) Analyze the algorithm used for edge tracking in Canny edge detection. (2)
- 3) Determine the significance of using standard databases and grand challenge datasets in the development and evaluation of medical image analysis algorithms. How do these datasets contribute to benchmarking performance, promoting reproducibility, and fostering collaboration in the medical imaging research community? (5)
 - A)
 - B) Generalize and discuss the ethical considerations related to the use of deep learning and AI technologies in medical image analysis. What are the potential risks and benefits of automated diagnosis, patient privacy concerns, and biases in algorithmic decision-making? (3)
 - C) Interpret whether the following statement is true or false and justify your answer. "*Root Mean Square contrast does not differentiate well between different intensity distributions*". (2)
- 4) A medical imaging startup aims to develop a cloud-based platform for automated segmentation of cardiac structures from cardiac MRI scans. The platform will be used by cardiologists for diagnosis, surgical planning, and treatment monitoring of (5)

- cardiovascular diseases. Given the scenario, determine a comprehensive approach for developing and deploying an automated segmentation algorithm for cardiac structures from MRI scans.
- A) (3)
- B) For the scenario given below, determine the preprocessing steps that you would adopt to reduce noise and enhance image contrast.
A medical imaging startup aims to develop a cloud-based platform for automated segmentation of cardiac structures from cardiac MRI scans. The platform will be used by cardiologists for diagnosis, surgical planning, and treatment monitoring of cardiovascular diseases. (3)
- C) A medical imaging startup aims to develop a medical image analysis tool that can segment livers from CT scans. However, they have decided to apply sobel filter to detect edges. Consider the following input image matrix :

45	30	23	28
80	156	180	180
180	47	31	25
54	56	57	80

(2)

Demonstrate how to apply the following sobel filter to the above sample image.

-1	0	1
-2	0	2
-1	0	1

- 5) Design a hypothetical scenario where DICOM plays a pivotal role in facilitating medical diagnosis or treatment planning and explain the technical properties of Medical Images. (5)
- A)
- B) Analyze the role of information systems in a hospital with the help of a neat diagram. (3)
- C) Explain the DICOM network management services in detail. (2)

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